

Makhura Kamogelo

Practicals week 10

Tasks:

Step 1: Diagnose and Fix

1. Work through: show ip interface brief, show running-config, show vlan brief, show ip route, show cdp neighbors.

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is visible with a central 2911 router connected to various devices including a Server, PCs, and a laptop. The right pane shows the CLI of the 2911 router. The CLI output shows the router is ready for configuration and displays the output of the 'show ip interface brief' command.

```
R1>enable
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/1	10.0.0.1	YES	manual	up	up
GigabitEthernet0/2	192.168.100.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Cisco Packet Tracer - C:\Users\Kamogelo Skamza\Downloads\Bradley Ramaphakela 1.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 427, y: 471

Time: 01:05:18

Search for device

Automatically Choose Connection Type

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1#
R1#show running-config
Building configuration...

Current configuration : 2309 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
enable secret 5 $1$mERr#Pq3/lr0agnISq0fab9TUVZ0
!
!
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
!
ip dhcp pool SALES
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
ip dhcp pool HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
ip dhcp pool IT
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8
!
!
ip cef
no ipv6 cef
```

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 427, y: 471

Time: 01:05:23

Search for device

Automatically Choose Connection Type

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1#
username admin secret 5 $1$mERr#Pq3/lr0agnISq0fab9TUVZ0
!
license udi pid CISCO02911/K9 an FTX152429H7-
!
!
no ip domain-lookup
ip domain-name net621.local
!
spanning-tree mode pvst
!
!
interface GigabitEthernet0/0
no ip address
ip nat inside
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
ip nat inside
```

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 427, y: 471

Time: 01:05:30

Search for device

Automatically Choose Connection Type

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
interface GigabitEthernet0/1
ip address 10.0.0.1 255.255.255.252
ip nat outside
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 192.168.100.1 255.255.255.0
ip nat inside
ip access-group 100 in
ip access-group 10 out
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
!
ip nat inside source list 1 interface GigabitEthernet0/1 overload
ip classless
ip route 192.168.80.0 255.255.255.0 10.0.0.2
!
ip flow-export version 9
!
access-list 100 permit tcp 192.168.20.0 0.0.0.255 host 192.168.100.10 eq vncv
access-list 100 permit tcp 192.168.10.0 0.0.0.255 host 192.168.100.10
access-list 100 deny ip any host 192.168.100.10
access-list 100 permit ip any any
access-list 1 permit 192.168.0.0 0.0.255.255
!
```

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 819, y: 63

Time: 01:05:59

Search for device

Automatically Choose Connection Type

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
access-list 100 permit tcp 192.168.20.0 0.0.0.255 host 192.168.100.10 eq vncv
access-list 100 permit tcp 192.168.10.0 0.0.0.255 host 192.168.100.10
access-list 100 deny ip any host 192.168.100.10
access-list 100 permit ip any any
access-list 1 permit 192.168.0.0 0.0.255.255
!
!
line con 0
!
line aux 0
!
line vty 0 4
login local
transport input ssh
!
end
R1#
R1#
R1#
R1#show vlan brief

```

VLAN Name	Status	Ports
1 default	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 859, y: 487

Time: 01:06:28

Search for device

Automatically Choose Connection Type

Physical Config CLI Attributes

IOS Command Line Interface

```
R1#
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, S - SGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

8.0.0.0/24 is subnetted, 1 subnets
O E2 8.8.8.0/24 [110/20] via 10.0.0.2, 00:03:04, GigabitEthernet0/1
C 10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
L 10.0.0.1/32 is directly connected, GigabitEthernet0/1
O IA 10.0.1.0/30 [110/21] via 10.0.0.2, 00:26:37, GigabitEthernet0/1
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
O 172.16.1.0/30 [110/1001] via 10.0.0.2, 00:26:37, GigabitEthernet0/1
O IA 172.16.2.1/32 [110/3] via 10.0.0.2, 00:26:37, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/10
L 192.168.10.1/32 is directly connected, GigabitEthernet0/10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, GigabitEthernet0/20
L 192.168.20.1/32 is directly connected, GigabitEthernet0/20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.30.0/24 is directly connected, GigabitEthernet0/30
L 192.168.30.1/32 is directly connected, GigabitEthernet0/30
S 192.168.50.0/24 [1/0] via 10.0.0.2
192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.100.0/24 is directly connected, GigabitEthernet0/2
L 192.168.100.1/32 is directly connected, GigabitEthernet0/2
```

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 823, y: 344

Time: 01:06:43

Search for device

Automatically Choose Connection Type

Physical Config CLI Attributes

IOS Command Line Interface

```
R1#
R1#
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, S - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, x - Repeater, P - Phone
Device ID Local Intrfce Holdtime Capability Platform Port ID
R-EDGE Gig 0/1 131 R C2900 Gig 0/0
R1#
R1#
R1#
```

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2. Document each fault found and the fix you applied (e.g. no shutdown, correcting a network statement, moving a port back to the right VLAN).

The image displays two screenshots of a Cisco Packet Tracer workspace, showing a network topology and a laptop's configuration window.

Top Screenshot:

- Network Topology:** A central 2911 R1 router is connected to a 2911 R-EDGE router. R1 is also connected to two 2911 R1 routers. Below R1, there are two 2911 R1 routers connected to a 2911 R1 router. The network includes several PCs (PC-SALES, PC-HR, PC-IT) and a Laptop-PT (CTU laptop). A Wireless-N Notebook Adapter is connected to the laptop.
- CTU Laptop Configuration:** The 'Desktop' tab is selected. The 'Wireless Network Name' is set to 'CORP-WIFI-V30'. The 'Site Information' section shows 'Wireless Mode' as 'Infrastructure', 'Wide Channel' as 'Auto', 'Standard Channel' as 'Auto', and 'Security' as 'WPA2-Personal'. The 'Authentication' is set to 'Auto'. The 'Connect' button is highlighted. The adapter is labeled 'Wireless-N Notebook Adapter' and 'Wireless Network Monitor v1.0'.

Bottom Screenshot:

- Network Topology:** Similar to the top screenshot, but the network configuration is slightly different, with the 2911 R1 router connected to the 2911 R-EDGE router. The network includes several PCs (PC-SALES, PC-HR, PC-IT) and a Laptop-PT (CTU laptop). A Wireless-N Notebook Adapter is connected to the laptop.
- CTU Laptop Configuration:** The 'Desktop' tab is selected. The 'Wireless Network Name' is set to 'Default'. The 'Site Information' section shows 'Wireless Mode' as 'Infrastructure', 'Wide Channel' as 'Auto', 'Standard Channel' as 'Auto', and 'Security' as 'WPA2-Personal'. The 'Authentication' is set to 'Auto'. The 'Connect' button is highlighted. The adapter is labeled 'Wireless-N Notebook Adapter' and 'Wireless Network Monitor v1.0'.

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 777, y: 408

Time: 00:06:34

Search for device

PT8200 4331 4321 JRB340 1941 2911

2911

CTU Laptop

Physical Config Desktop Programming Attributes

Link Information Connect Profiles

To connect to a network, select the profile name then click the Connect button. To create or edit a profile, use the menu bar at the bottom of the screen.

Profile	Wireless Network Name
CORP-WIFI-V30	CORP-WIFI-V30
Default	Default

Site Information

Wireless Mode: Infrastructure
Wide Channel: Auto
Standard Channel: Auto
Security: WPA2-Personal
Authentication: Auto

Connect

2.4GHz

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. WPC300N

New Edit Import Export Delete Adapter is Active

Simulation

Periodic Num

17°C Partly cloudy 7:32 PM 10/4/2026

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 237, y: 109

Time: 00:28:57

Search for device

Automatically Choose Connection Type

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
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Realtime Simulation

17°C Partly cloudy 7:53 PM 10/4/2026

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File Edit Options View Tools Extensions Window Help

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.1

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::200:CFF:FE3D:40AA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Time: 00:27:31

Search for device

17°C Mostly clear 7:54 PM 10/4/2026

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File Edit Options View Tools Extensions Window Help

Physical Config CLI Attributes

IOS Command Line Interface

Enter configuration commands, one per line. End with CNTL/Z.

```
R1(config)#interface g0/0
R1(config-if)#shutdown

R1(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to down
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to down
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to down
R1(config-if)#interface g0/1
R1(config-if)#shutdown

R1(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
13:24:09: %OSPF-5-ADJCHG: Process 1, Nbr 198.51.100.1 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down: Interface down or detached
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#
```

Time: 00:37:55

Search for device

17°C Mostly clear 8:39 PM 10/4/2026

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 835, y: 290

Time: 00:39:07

Search for device

Automatically Choose Connection Type

Physical Config CLI Attributes

IOS Command Line Interface

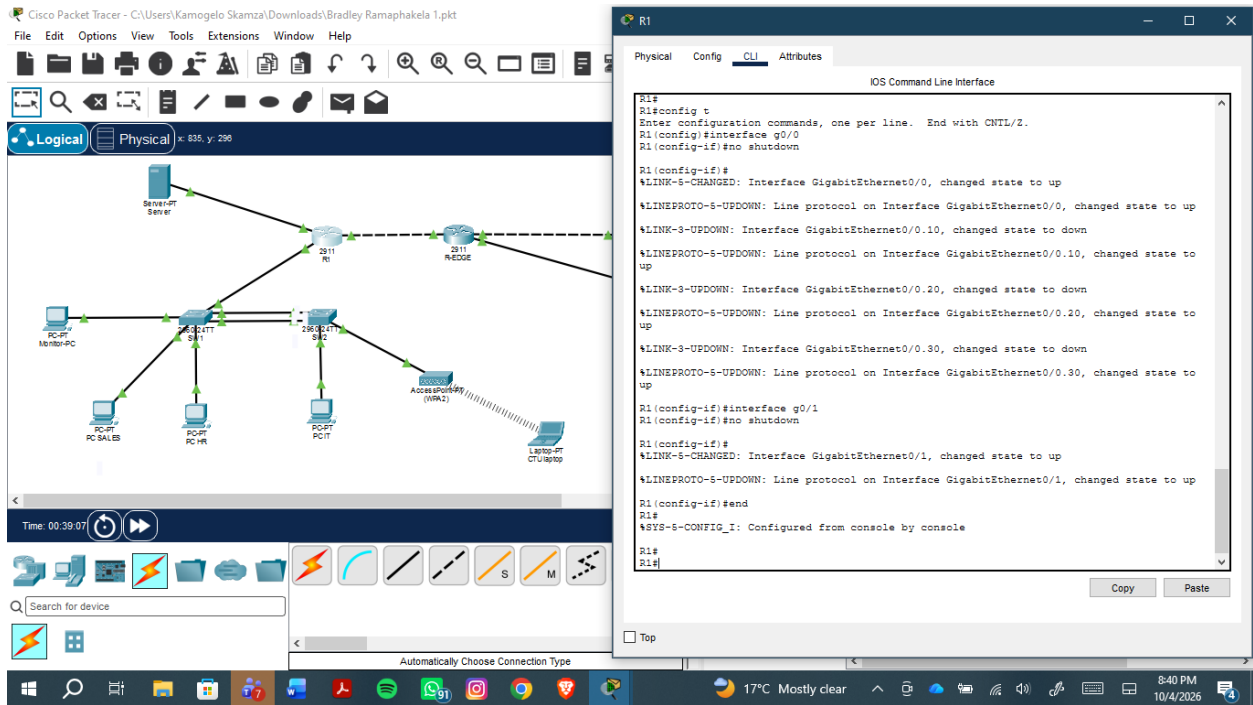
```
R1#
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
R1(config-if)#interface g0/1
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R1(config-if)#end
R1#
R1#
R1#
```

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Top



Reflection Questions (answer these in your submission):

1. Of the faults you found, which was hardest to diagnose, and which command finally pointed you to it?

The hardest fault to diagnose was the incorrect interface configuration because the interface appeared to be connected, but it was not communicating correctly. The command that finally pointed me to the problem was `show ip interface brief`. This command showed the status of the interfaces and helped identify the interface that was down or had an incorrect IP configuration. After correcting the interface configuration, connectivity was restored.